SEMINAR III · WEEK 5

Percent: What Part of the Whole?

Percent means out of 100. Once you can see a percent as a fraction and a decimal wearing a different outfit, discounts, tax, tips, and percent change all become the same handful of moves.

What Does Percent Mean?

25% means 25 out of 100 — the same amount as 25/100, 0.25, or 1/4. All four are the same quantity, just written differently.

Fraction, Decimal, or Percent?

The representation changes. The quantity doesn't. You'll move between all three constantly, so these benchmarks are worth knowing cold:

10% = 1/10 | 25% = 1/4 | 50% = 1/2 | 75% = 3/4

Percent Shortcuts Worth Knowing

You can find most percentages in your head by combining a few benchmarks: 1%, 5%, 10%, 20%, 25%, 50%, 75%, 100%.

Example: 15% of a number = 10% of it, plus 5% of it (which is half of the 10%).

Which Percent Problem Is This?

Every percent problem is one of three types. Before calculating, name which one you have.

Type	You know...	You're finding...

Find the part	the percent and the whole	an amount
Find the percent	the part and the whole	a percentage
Find the whole	the part and the percent	the total

Finding a Percent of a Number

30% of 80: write 30% as 0.30, then multiply: 0.30 × 80 = **24**.

Finding What Percent One Number Is of Another

Divide the part by the whole, then convert the decimal to a percent.

18 out of 24 questions correct: 18 ÷ 24 = 0.75 = **75%**.

Finding the Whole

Try benchmark reasoning first — if you know a smaller piece is a familiar percent, you may not need to calculate at all. Otherwise, divide the part by the decimal percent.

15 is 20% of what number? 15 ÷ 0.20 = **75**. Check: 20% of 75 is 15. ✓

Discounts

Two steps: calculate the discount, then subtract it from the original price. Keep the **discount amount** and the **sale price** straight — they're not the same number.

An $80 jacket, 25% off: discount = 0.25×80 = $20. Sale price = 80−20 = **$60**.

Tax

Two steps: calculate the tax, then add it to the original price. Keep the **tax amount** and the **final cost** straight.

> **A $50 purchase, 8% tax:** tax = 0.08×50 = $4. Final cost = 50+4 = **$54**.

Tips

You don't need exact restaurant-etiquette rules — you need a fast, useful estimate. Find 10% by moving the decimal point one place, double it for 20%, or add half of the 10% for 15%.

Percent Increase and Decrease

Five steps: identify the original value, identify the new value, find the change, divide the change by the **original** value, convert to a percent.

> **A price rises from 40 to 50:** change = 10. 10 ÷ 40 = 0.25 = **25% increase**.

Does My Answer Make Sense?

- Is the part smaller than the whole when the percent is under 100%?
- Should a discount decrease the price? Should tax increase it?
- Did I use the *original* value for percent change?
- Is my answer near what I estimated?
- Did I answer with the percent *amount*, or the final amount — whichever the question actually asked for?

Error Detective

> **Using 20 instead of 0.20**

A student calculates 20 × 80 instead of 0.20 × 80 — treating the percent as a whole number rather than converting it first. Fix: convert to a decimal (or fraction) before multiplying.

Confusing discount with sale price

A student correctly finds the $20 discount on an $80 item, then reports $20 as the sale price. The calculation was right; the question asked for something else. Fix: identify exactly what's being asked before answering.

Reversing part and whole

18 out of 24 becomes 24 ÷ 18 instead of 18 ÷ 24. Fix: identify which number is the part and which is the whole before dividing.

Using the new value for percent change

A price rises from 40 to 50, and a student divides the change by 50 instead of 40. Fix: percent change always divides by the **original** value.

Quick Reference

Meaning: percent = out of 100.

Convert: percent → decimal, divide by 100. Decimal → percent, multiply by 100.

Benchmarks: 10%=1/10, 25%=1/4, 50%=1/2, 75%=3/4.

Three questions: find the part, find the percent, find the whole.

Money: discount → subtract. Tax → add.

Percent change: change ÷ original.

Check: Operation → Size → Direction → Unit → Context.

Keep this page. You'll use it all year.